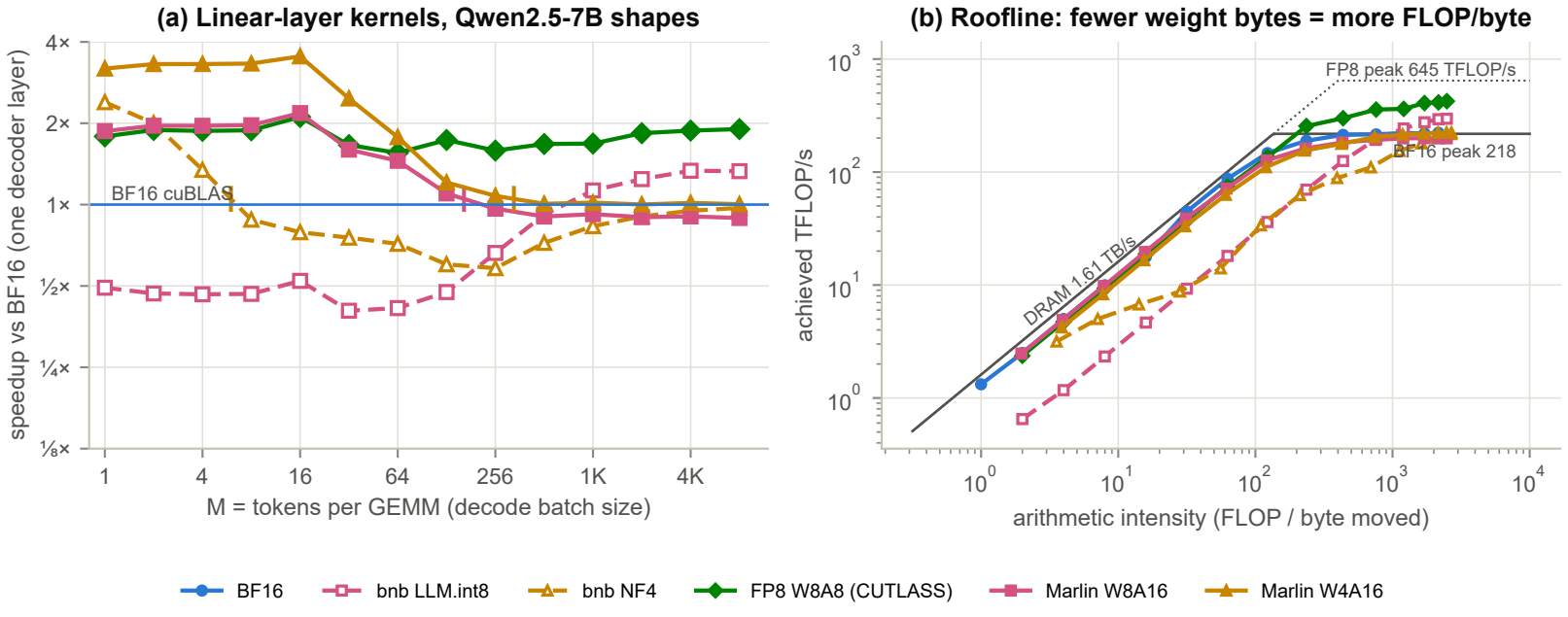

(a) Linear-layer kernels, Qwen2.5-7B shapes
speedup vs BF16 (one decoder layer)
4×
2×
1×
½×
¼×
⅛×
BF16 cuBLAS
1
4
16
64
256
1K
4K
M = tokens per GEMM (decode batch size)
(b) Roofline: fewer weight bytes = more FLOP/byte
achieved TFLOP/s
$10^3$
$10^2$
$10^1$
$10^0$
FP8 peak 645 TFLOP/s
BF16 peak 218
DRAM 1.61 TB/s
$10^0$
$10^1$
$10^2$
$10^3$
$10^4$
arithmetic intensity (FLOP / byte moved)
BF16
bnb LLM.int8
bnb NF4
FP8 W8A8 (CUTLASS)
Marlin W8A16
Marlin W4A16